RAMDEOBABA UNIVERSITY, NAGPUR

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
B.Tech. CSE (Artificial Intelligence and Data Science)

SEMESTER V
STATISTICAL INFERENCE LAB
PROJECT REPORT

STUDYSIGNAL

An Interactive Statistical Inference System for Discovering the Hidden Factors Behind Student Performance

Submitted by

Jay Raut
Roll No.: 35
Enrollment No.: _____

Project Guide: _____

Academic Year: 2026–27

StudySignal — Statistical Inference Lab Project Report

1. PROJECT OBJECTIVE

StudySignal demonstrates how statistical inference can be used to investigate factors associated with student academic performance. It explores a structured dataset, summarizes distributions, estimates a confidence interval for average marks, tests relationships between variables, compares group means, and presents interpretations in an interactive dashboard. The project uses synthetic data and is intended for education, not for making high-stakes decisions about real students.

Objectives: generate and inspect a reproducible dataset; calculate descriptive statistics; quantify numeric associations with Pearson and Spearman correlation; compare two independent groups with Welch's t-test; compare three study-method groups using one-way ANOVA; test categorical association using chi-square; estimate a 95% confidence interval; and provide an interactive dashboard.

2. INTRODUCTION

Academic performance may be associated with study time, sleep, attendance, screen time, stress, and study method. Charts describe patterns, but inferential statistics help quantify uncertainty and test whether observed differences are inconsistent with a specified null hypothesis. StudySignal integrates these techniques into one workflow and explains their interpretations and limitations.

2.1 Problem Statement

Students and educators may observe patterns in academic data without knowing whether they are statistically meaningful. StudySignal provides a transparent and repeatable workflow for selecting an appropriate statistical method, running the test, and interpreting the result.

2.2 Scope of the Project

Included: synthetic data generation, data inspection, descriptive summaries, correlation and covariance, Pearson and Spearman tests, Welch independent-samples t-test, one-way ANOVA, chi-square test, confidence intervals, exploratory linear regression, visualizations, and a Streamlit dashboard. Excluded: real student records, causal claims, clinical or psychological assessment, and institutional decision-making.

3. PROJECT METHODOLOGY

The workflow consists of data generation, quality checks, exploratory analysis, inferential testing, interpretation, and presentation through a notebook and interactive dashboard.

3.1 Dataset Loading and Exploration

Dataset: StudySignal Student Performance Dataset. Source: generated by project code with a fixed random seed; no real-person records are used. The CSV contains 300 synthetic records and 10 columns: Student_ID, Study_Hours, Sleep_Hours, Attendance_Percent, Screen_Time_Hours, Stress_Score, Study_Method, Group_Study, Marks, and Pass_Status. The notebook displays sample rows, data types, descriptive statistics, missing-value counts, duplicate count, and pass-status distribution.

The generated dataset has 300 records, a mean mark of 62.76, median of 62.85, and a simulated pass rate of 98.7%. These are outputs from synthetic data, not real institutional statistics.

3.2 Data Preprocessing

Numeric variables are generated within bounded ranges. Marks are restricted to 0–100. Pass_Status is derived from Marks using a threshold of 40. The notebook checks for missing values and duplicates. Categorical labels are retained for group comparisons and contingency tables. The exploratory regression uses Study_Hours, Sleep_Hours, Attendance_Percent, Screen_Time_Hours, and Stress_Score as predictors and Marks as the response, with an 80:20 train-test split and fixed random state.

3.3 Feature Extraction / Selection

No dimensionality-reduction method is required. The selected variables correspond to the research questions. An attendance band (low, medium, high) is derived for the chi-square test, and Pass_Status is derived from the marks threshold. All transformations are explicitly defined in code.

3.4 Statistical Tests and Analytical Methods

Sr. No.	Method	Purpose / Notes
1	Pearson correlation	Measures linear association between study hours and marks and provides a p-value.
2	Spearman correlation	Measures monotonic rank association; useful as a rank-based comparison.
3	Welch independent t-test	Compares group-study and solo-study mean marks without assuming equal variances.
4	One-way ANOVA	Tests whether mean marks are equal across three study methods.
5	Chi-square test	Tests association between attendance band and pass status.
6	95% t confidence interval	Estimates an interval for the population mean under sampling assumptions.
7	Linear regression	Provides an exploratory prediction baseline evaluated with MAE, RMSE, and R^2.

3.5 Performance Evaluation and Result Summary

For hypothesis tests, the significance threshold is $\alpha = 0.05$. A p-value below this threshold is treated as evidence against the specified null hypothesis. Regression error metrics are reported separately.

Analysis	Statistic	p-value / estimate	Interpretation
Pearson: study hours vs marks	$r = 0.593$	7.12e-30	Significant association
Spearman: study hours vs marks	$\rho = 0.558$	6.05e-26	Significant association
Welch t-test: group vs solo	$t = 3.313$	0.00116	Significant mean difference
One-way ANOVA	$F = 6.520$	0.00169	At least one mean differs
Chi-square: attendance vs pass status	$\chi^2 = 1.778$	0.411	Not significant
Mean marks, 95% CI	62.76	(61.58, 63.95)	Interval estimate
Regression on test set	$R^2 = 0.525$	MAE = 5.86; RMSE = 7.53	Exploratory prediction baseline

Values are generated from the included seeded synthetic dataset. Rerun the notebook if the data-generation process is changed.

3.6 Best Method Selection

This project does not select one universal best model. Each method answers a different question: correlation quantifies association, t-tests and ANOVA compare means, chi-square tests categorical association, confidence intervals quantify uncertainty, and regression summarizes predictive performance. The method is selected according to variable type and research question.

4. PROJECT IMPLEMENTATION

4.1 Tools and Technologies

Tool / Technology	Version	Purpose
Python	3.x	Programming language
Jupyter / Google Colab	Notebook environment	Reproducible analysis

Tool / Technology	Version	Purpose
Pandas / NumPy	Environment versions	Data handling
SciPy	Environment version	Inferential tests
Matplotlib	Environment version	Visualization
Scikit-learn	Environment version	Regression and evaluation
Streamlit	Environment version	Interactive dashboard

4.2 Module-wise Implementation

1. Dataset generation and export. 2. Data quality and exploration. 3. Visualization of numeric relationships and group distributions. 4. Inference engine for correlation, t-test, ANOVA, chi-square, and confidence intervals. 5. Regression module reporting MAE, RMSE, and R^2. 6. Dashboard for interactive variable selection and statistical output.

4.3 GUI Implementation

The Streamlit interface in app.py displays summary metrics, dataset preview, selectable numeric variable and correlation method, scatter plot, one-way ANOVA results for study methods, attendance/pass-status contingency table, chi-square result, and a 95% confidence interval.

Launch instructions: install Python 3.10 or later, run `pip install -r requirements.txt`, then run `streamlit run app.py` from the project directory. The notebook can be opened in Jupyter or uploaded to Google Colab.

5. COMPLETE CODE

Complete runnable analysis: StudySignal_Statistical_Inference.ipynb. Interactive interface: app.py. Dataset: studysignal_student_dataset.csv. The notebook generates its own dataset, so it does not require an uploaded file. Keep the files together in the project repository.

6. RESULTS AND DISCUSSION

The analysis demonstrates that different inferential methods answer different questions. Correlation evaluates association between numeric variables; Welch’s t-test compares two independent means; one-way ANOVA compares three means; chi-square evaluates categorical association; and a confidence interval quantifies uncertainty around the sample mean. Regression metrics describe prediction error and are not hypothesis-test p-values.

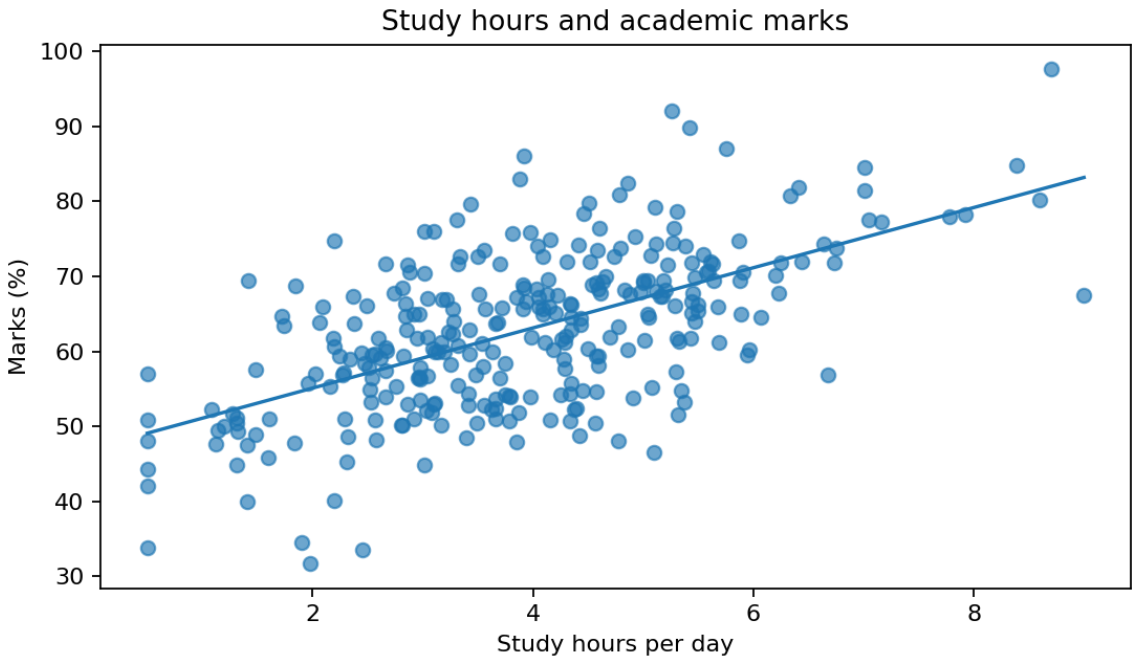


Figure 1. Study hours and marks in the synthetic dataset. The fitted line summarizes the observed linear trend and does not establish causation.

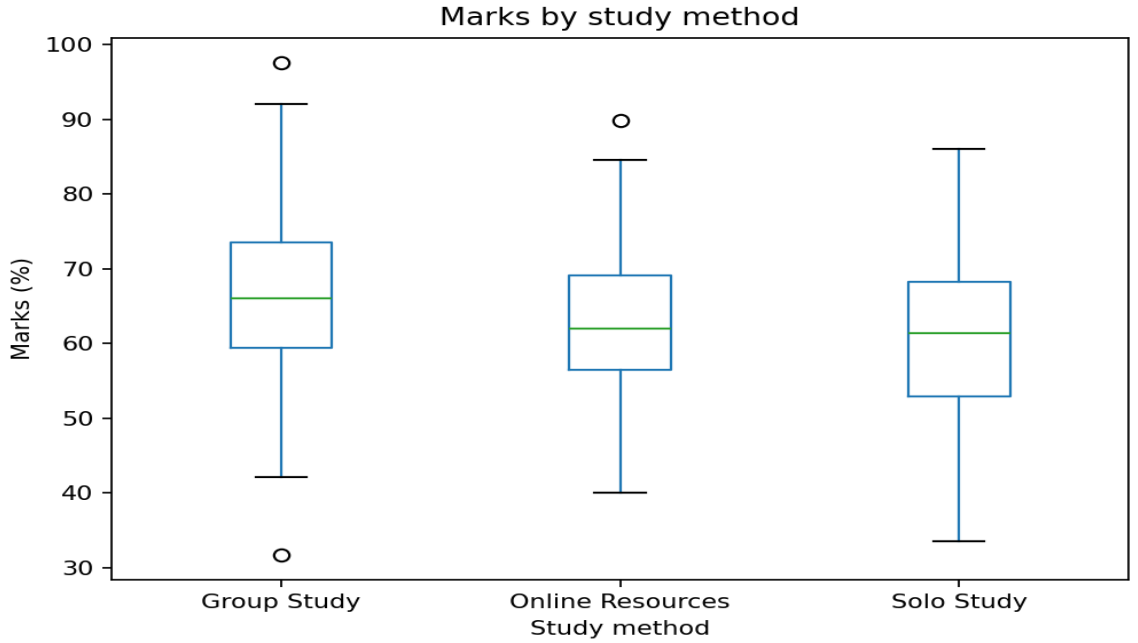


Figure 2. Distribution of marks across study methods. ANOVA tests whether the group means can be treated as equal.

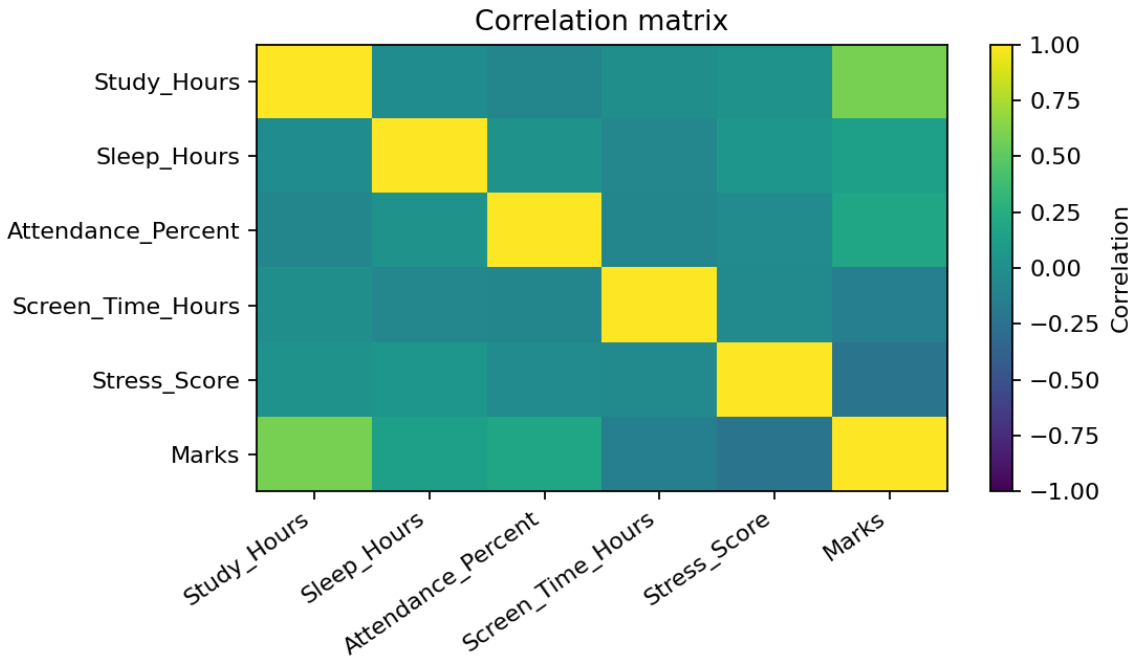


Figure 3. Correlation matrix for selected numeric variables. Correlation is pairwise association and may reflect other variables or simulation assumptions.

6.1 Required Output Screenshots

The report includes analysis figures generated by the project. For final submission, capture actual notebook outputs showing preprocessing, the statistical result summary, correlation plot, t-test, ANOVA, chi-square, confidence interval, regression metrics, and the running Streamlit interface with example outputs. Screenshots should be captured from the executed notebook/app.

6.2 Result Analysis

The sample mean is 62.76, with a 95% confidence interval of (61.58, 63.95). Pearson correlation between study hours and marks is 0.593 (p = 7.12e-30). The Welch test compares group-study and solo-study records; ANOVA compares three study methods; chi-square examines attendance band and pass status.

Limitations: the data are simulated and the relationships are partly specified by the generation formula. Therefore, results cannot support conclusions about real students. Statistical significance depends on assumptions, sample design, and data quality. The analysis does not establish causation, and the regression model should not predict actual grades.

7. CONCLUSION

StudySignal integrates descriptive analysis, correlation, hypothesis testing, confidence intervals, ANOVA, chi-square testing, and a simple regression baseline into one reproducible workflow. Its main contribution is to connect each research question to a suitable method and explain what can and cannot be concluded. Future improvements include consent-based anonymized data, assumption diagnostics, effect sizes, post-hoc tests, and CSV upload support.

8. GITHUB LINK

GitHub Repository URL: To be added after creating and uploading the repository. Recommended contents: notebook, app.py, dataset or generation code, requirements.txt, README.md, report PDF, and figures. Do not enter a repository URL until the repository exists.

9. REFERENCES

[1] SciPy documentation, Statistical functions: https://docs.scipy.org/doc/scipy/reference/stats.html

[2] pandas documentation: https://pandas.pydata.org/docs/

[3] NumPy documentation: https://numpy.org/doc/

[4] Matplotlib documentation: https://matplotlib.org/stable/

[5] scikit-learn documentation: https://scikit-learn.org/stable/

[6] Streamlit documentation: https://docs.streamlit.io/

[7] StudySignal Student Performance Dataset: generated by the project code; synthetic educational data.

FINAL SUBMISSION CHECKLIST

■ Project report PDF. ■ Runnable notebook (.ipynb). ■ Notebook exported as PDF after execution. ■ GitHub repository with code, dataset/generation code, requirements, README, and report. ■ Actual screenshots captured from the executed notebook and dashboard. ■ Enrollment number and guide details completed on the cover. ■ GitHub URL added after repository creation.